Step 5 : Calculation of test statistic

To find $\bar{d}$ and s:

Let x denote sales before training and y denote sales after training

Salesmen	x_i	y_i	$d_i = y_i - x_i$	$d_i - \bar{d}$	$(d_i - \bar{d})^2$
1	15	17	2	0	0
2	22	23	1	-1	1
3	6	16	10	8	64
4	17	20	3	1	1
5	12	14	2	0	0
6	20	21	1	-1	1
7	18	18	0	-2	4
8	14	20	6	4	16
9	10	10	0	-2	4
10	16	11	-5	-7	49
		Total	$\sum_{i=1}^{n} d_i = 20$	$\sum_{i=1}^{n} (d_i - \bar{d}) = 0$	$\sum_{i=1}^{n} (d_i - \bar{d})^2 = 140$

Here instead of $d_i = x_i - y_i$ it is assumed $d_i = y_i - x_i$ for calculations to be simpler.

$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n} = \frac{20}{10} = 2$$

$$s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^{n} (d_i - \bar{d})^2} = \sqrt{\frac{140}{9}} = \sqrt{15.56} = 3.94$$

The calculated value of the statistic is

$$t_0 = \frac{\bar{d}}{\frac{s}{\sqrt{n}}} = \frac{2}{\frac{3.94}{\sqrt{10}}} = 1.6052$$

Step 6 : Critical value

Since H_0 is a one-sided alternative hypothesis, the critical value at 5% level of significance is $t_e = t_{n-1,\,\alpha} = t_{9,0.05} = 1.833$

Step 7 : Decision

It is a one-tailed test. Since $|t_0| = 1.6052 < t_e = t_{n-1,\,\alpha} = t_{9,0.05} = 1.833$, H_0 is not rejected. Hence, there is no evidence that the mean sales has increased after the training.

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